

Math 142, F19
Quiz 4

Name: Answer

Instructions: You must show all of your work to receive credits for correct answers.

Problem 1. (5 points) Find $\int \frac{1}{x^2 \sqrt{16-x^2}} dx$. (Hint: using the substitution $x = 4 \sin \theta$).

Let $x = 4 \sin \theta$, then $dx = 4 \cos \theta d\theta$

$$\int \frac{1}{x^2 \sqrt{16-x^2}} dx = \int \frac{1}{(4 \sin \theta)^2 \sqrt{16-(4 \sin \theta)^2}} \cdot 4 \cos \theta d\theta$$

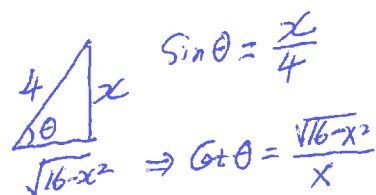
$$= \int \frac{1}{16 \sin^2 \theta \cdot 4 \cos \theta} \cdot 4 \cos \theta d\theta$$

$$= \int \frac{1}{16 \sin^2 \theta} d\theta$$

$$= \int \frac{1}{16} \csc^2 \theta d\theta$$

$$= -\frac{1}{16} \cot \theta + C$$

$$= -\frac{1}{16} \frac{\sqrt{16-x^2}}{x} + C$$



$\sin \theta = \frac{x}{4}$
 $\cot \theta = \frac{\sqrt{16-x^2}}{x}$

Problem 2. (5 points) Evaluate $\int \frac{x-13}{x^2-2x-15} dx$ using the partial fraction decomposition.

$$\frac{x-13}{x^2-2x-15} = \frac{x-13}{(x-5)(x+3)} = \frac{A}{x-5} + \frac{B}{x+3} = \frac{A(x+3)+B(x-5)}{(x-5)(x+3)}$$

$$= \frac{Ax+3A+Bx-5B}{(x-5)(x+3)} = \frac{(A+B)x+(3A-5B)}{(x-5)(x+3)}$$

$$\Rightarrow \begin{cases} A+B=1 \\ 3A-5B=-13 \end{cases} \Rightarrow \begin{cases} A=-1 \\ B=2 \end{cases}$$

$$\text{Thus } \int \frac{x-13}{x^2-2x-15} dx = \int -\frac{1}{x-5} + \frac{2}{x+3} dx$$

$$= -\ln|x-5| + 2\ln|x+3| + C$$